

layout: title

title: Iterative Methods

subtitle: Gradient descent, Newton, and search

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date: 2026

Iterative Methods

layout: section

section_number: 1

title: Optimization

subtitle: First-order and second-order updates

Optimization

layout: two-column
title: Gradient Descent

Gradient Descent

We want to minimize a differentiable objective.

$$f(x) = x^2$$

The update follows the negative gradient:

$$x_{n+1} = x_n - \eta \nabla f(x_n)$$

```
gradient_descent(f=x**2, x=1.5)
```

Move in the direction of the negative gradient.

layout: equation-focus
title: Newton's Method

Newton's Method

Newton's method iteratively improves an estimate of a root.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

For a square root of two, solve ($f(x) = x^2 - 2$).

layout: diagram-focus
title: Newton on $x^2 - 2$

Newton on $x^2 - 2$

```
:::diagram
newton_method(f=x**2-2, x0=1.5)
:::
```

Each tangent hits the axis at the next iterate.

layout: title-content

title: Chain Rule

Chain Rule

If ($y = f(g(x))$), differentiation composes linear maps:

$$\frac{dy}{dx} = f'(g(x)) g'(x)$$

- Outer derivative evaluated at the inner value
- Inner derivative scales the infinitesimal

layout: code-focus
title: Binary Search

Binary Search

Halve a sorted range until the target is found.

```
def binary_search(arr, target):  
    lo, hi = 0, len(arr) - 1  
    while lo <= hi:  
        mid = (lo + hi) // 2  
        if arr[mid] == target:  
            return mid  
        if arr[mid] < target:  
            lo = mid + 1  
        else:  
            hi = mid - 1  
    return -1
```